# The Comparative Measurement of Welfare in Broiler Chickens: Assessing the Impact of Reforms

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Since the 1960s, profound transformations in the livestock industry took place, but nowhere else were they as dramatic as in the poultry industry. This was particularly pronounced in the production of birds raised for their meat (known as broiler chickens), which has expanded from six billion birds slaughtered per year in 1960 to over 70 billion in 2020. Chicken is now the most numerous land animal on the planet [1] and the most consumed animal protein by humans, with no major cultural or religious limitations on its consumption.

The ten-fold increase in broiler production in the recent past occurred alongside radical changes in their management, with the emergence of highly industrialized production processes involving the intensive production of large numbers of animals genetically selected for high productivity in factory-style farms [2–7]. These farms range from low-tech sheds where crowded animals are fattened, to sophisticated facilities where productivity is further boosted by nutrition and husbandry practices that ensure the highest possible growth and production in the shortest possible time and the lowest possible cost. Intensive or industrial food animal production systems [8] are responsible for the slaughter of most chickens at a global scale. These highly industrialized processes involve the production of birds at extremely high-stocking densities in barren factory-style facilities holding thousands of individuals. At the same time, intensive genetic selection for rapid muscle growth and higher feed conversion rapidly transformed their anatomy and physiology. Average daily weight gain has increased by over 400% in the last decades [9], with genetic selection directly causing most of this increase [10]. Pectoral mass now accounts for about 20% of body weight, twice as much as that of ancestral varieties. Age at slaughter was also reduced: from 16 to only 6 or fewer weeks — much heavier chickens are now slaughtered at much younger ages.

Intensive management and genetic selection for traits desirable for fast and cheap meat production has taken a heavy toll on the health and well-being of broilers [11,12]. Growing birds are now afflicted by a range of conditions, including poor cardiorespiratory and immune function, sudden death and ascites syndrome and a high incidence of lameness-causing conditions as a result of the high stress placed on relatively immature bones and joints, changes in body conformation, problems of ossification and infections. These conditions, in turn, cause chickens to spend extended periods of time sitting or lying on litter, which will often produce skin lesions. The expression of motivated behaviors has concomitantly become physiologically and physically challenging, and further limited

[1] Welfare Footprint Project (https://welfarefootprint.org/), Center for Welfare Metrics.

by the high stocking densities and a lack of environmental enrichment. Breeder birds (parents, grandparents and great-grandparents of meat chickens) experience even greater challenges: while sharing the same genetics for fast-growth, breeders must live long enough to reach adulthood and reproduce. To prevent them from becoming obese, severely ill or dying over this longer lifespan, severe feed restriction is commonly practiced.

Commercial broiler production has been widely recognized as one of the most serious animal welfare issues in the global agricultural sector [10,13–15]. Among consumers, ethical concerns about the quality of life of meat chickens are also rapidly growing. For example, in a 2018 survey conducted for the National Chicken Council in the United States, most respondents said they were concerned about how chickens are raised [16]. While there have been some improvements in management such as climate control and the emergence of some welfare certification schemes, severe welfare challenges remain that cannot be mitigated by a focus on management only. As a result, campaigns to improve the way broiler chickens are produced, which include the use of alternative genetic strains of slower-growing broiler chickens, have been gathering pace in many countries.

Recently there has been increasing pressure on producers and retailers to adopt the recommendations set out in the Better Chicken Commitment (BCC), a set of standards supported by a coalition of animal protection organizations, and which can be summarized in five tenets: (i) transitioning to higher welfare breeds (Hubbard JA 757, 787, 957, or 987, Rambler Ranger, Ranger Classic, and Ranger Gold, or others that meet the criteria of the RSPCA Broiler Breed Welfare Assessment Protocol), (2) reducing stocking density (maximum of 30 Kg/m2) and prohibiting cages, (3) provisioning of enriched environments (light intensity of at least 50 lux, at least two meters of useable perch space and two pecking substrates per 1,000 birds), (4) improving stunning and slaughter systems (replacing live-shackling by controlled-atmosphere processing systems that induce an irreversible stun) and (5) demonstrating compliance with the standards via third-party auditing. In addition, the use of thinning (i.e. the removal of a fraction of the flock prior to the final depopulation) is discouraged, but if practiced must be limited to one thinning per flock.

In this book, we use the Cumulative Pain Framework (CPF) [17], a novel analytical approach that enables quantifying welfare, to estimate the welfare of commercial broiler chickens in different production scenarios and the impact of adopting the BCC. We also investigate the potential welfare trade-offs associated with the use of slower growing breeds, including the extent to which reduced time in pain due to better genetics and improved environments (as reflected in the incidence and severity of key welfare challenges) will exceed a longer time of exposure to still existing welfare challenges due to the longer lifespan of slower-growing birds.

## Cumulative Time in Pain Experienced by Commercial Broiler Chickens

The CPF framework has already been successfully used to quantify the welfare of egg-laying hens in different indoor housing systems, and the impact of the transition to cage-free settings [18]. A detailed description of the methodological approach is available elsewhere [17]. Briefly, the CPF quantifies the cumulative load of negative affective experiences endured over any period of time using a meaningful metric: time in pain (where the term pain is used as a shorthand for both physical and psychological pain[2]). To this end, it translates evidence on the duration and intensity of the pain associated with each of multiple welfare challenges typically experienced by a population into total time spent in four categories of pain intensity: Annoying, Hurtful, Disabling and Excruciating (Box 1), an exercise grounded on observations of its manifestations (e.g. neurological, physiological, behavioral) and evolutionary reasoning.

## Box 1. Definition of Pain Intensity Categories

**Annoying**: experiences of pain perceived as aversive, but not intense enough to disrupt the animal's routine in a way that alters adaptive functioning or affects the behaviors that animals are motivated to perform. Similarly, Annoying pain should not deter individuals from enjoying pleasant experiences with no short-term function (e.g., play) and positive social interactions. Sufferers can ignore this sensation most of the time. Performance of cognitive tasks demanding attention are either not affected or only mildly affected. Physiological departures from expected baseline values are not expected to be present. Vocalizations and other overt expressions of pain should not be observed.

**Hurtful**: experiences in this category disrupt the ability of individuals to function optimally. Different from Annoying pain, the ability to draw attention away from the sensation of pain is reduced: awareness of pain is likely to be present most of the time, interspersed by brief periods during which pain can be ignored depending on the level of distraction provided by other activities. Individuals can still conduct routine activities that are important in the short-term (e.g. eating, foraging) and perform cognitively demanding tasks, but an impairment in their ability or motivation to do so is likely to be observed. Although animals may still engage in behaviors they are strongly motivated to perform (i.e., exploratory, comfort, sexual, and maintenance behaviors), their frequency or duration is likely to be reduced. Engagement in positive activities with no immediate benefits (e.g., play in piglets, dustbathing in chickens) is not expected. Reduced alertness and inattention to ongoing stimuli may be present. The effect of (effective) drugs (e.g., analgesics if pain is physical, psychotropic drugs in the case of psychological pain) in the alleviation of symptoms is expected.

**Disabling**: pain at this level takes priority over most bids for behavioral execution, and prevents all forms of enjoyment or positive welfare. Pain is continuously distressing. Individuals affected by harms in this category change their activity levels drastically (the degree of disruption in the ability of an organism to function optimally should not be confused with the overt expression of pain behaviors, which is less likely in prey species). Inattention and unresponsiveness to milder forms of pain or other ongoing stimuli and surroundings is likely to be observed. Relief often requires higher drug dosages or more powerful drugs. The term Disabling refers to the disability caused by 'pain', not to any structural disability.

**Excruciating**: all conditions associated with extreme levels of pain that are not normally tolerated even if only for a few seconds. In humans, it would mark the threshold of pain under which many people choose to take their life rather than endure the pain. This is the case, for example, of scalding and severe burning events. Behavioral patterns associated with experiences in this category may include loud screaming, involuntary shaking, extreme muscle tension or extreme restlessness. Another criterion is the manifestation of behaviors that individuals would strongly refrain from displaying under normal circumstances, as they threaten body integrity (e.g. running into hazardous areas or exposing oneself to sources of danger, such as predators, as a result of pain or of attempts to alleviate it). The attribution of conditions to this level must be done cautiously. Concealment of pain is not possible.

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[2] Pain is operationally defined as 'any felt negative affective state', encompassing negative affective experiences of both somatic origin (physical pain) and more related to the primary emotional systems (psychological pain).

By considering the prevalence of each welfare challenge in the population, estimated times in pain are also determined for the average population member. To describe and assess the intensity and duration of the pain caused by each welfare challenge, a standardized visual framework, which we refer to as Pain-Track [19], is used (Box 2). The method is transparent, brings important knowledge gaps into light and incorporates uncertainty about all parameters into the final estimates. Importantly, since the method is based on a universal metric at the heart of how sentient organisms experience life (time), it enables comparing and combining the impact of experiences of very different nature.

## Box 2. Pain-Tracks

In a Pain-Track, the four categories of pain intensity are vertically ordered. Time is segmented in columns, where each segment is used to characterize a fairly homogeneous experience of the pain. Since pain experiences can unfold over a wide range of periods (from milliseconds to years), time segments can be associated with different time units to ensure flexibility.

In each time segment, cells can be filled with either (i) the probability that the intensity of the pain belongs to each of the four categories (capturing situations where there is uncertainty in the classification of pain intensity) or (ii) the estimated proportion of the population that experiences pain at each intensity (capturing variability in how different individuals may experience pain from a challenge). A fifth row ("no pain") can be also included. The duration of each segment is also represented by an uncertainty range.

An example is provided below, showing a hypothesis for the experience of pain associated with an untreated skin wound in a bird. There are four segments: (i) the moment of wound infliction; (ii) the period of hemostasis (coagulation); (iii) the acute inflammatory phase; and (iv) the proliferative (repair) phase, when residual inflammation (hence pain) may still be present. In each segment, the probability that the pain belongs to each intensity category is assigned. At the population level, the percentages may also be interpreted as the proportion of the population that experiences pain at each intensity. A detailed justification of the values shown in this Pain-Track is available elsewhere [20].

Each value displayed in a Pain-Track should be based on a comprehensive review of knowledge from different disciplines and sources. In addition to organizing and summarizing existing evidence, Pain-Tracks require that assumptions are made explicit, thus facilitating scrutiny and debate. Because evidence used to fill in the Pain-Track will be often limited, criticism of the proposed values is encouraged.

| | i.<br>Wound infliction<br>Point of tissue rupture | ii.<br>Hemostasis<br>(coagulation) | iii.<br>Acute<br>inflammation | iv.<br>Inflammation +<br>proliferation |
|---|---|---|---|---|
| **Excruciating** | | | | |
| **Disabling** | 60% | 20% | | |
| **Hurtful** | 40% | 80% | 70% | 20% |
| **Annoying** | | | 30% | 60% |
| **Duration** | **0.5 - 2 min** | **5 - 15 min** | **0.5 - 2 days** | **7 - 14 days** |

## Conventional and Reformed Scenarios

We analyze the following scenarios, for which data on broiler welfare was available. First, a conventional scenario represented by the use of fast-growing breeds (e.g., Aviagen Ross 308, 708, Cobb 500). Although the performance objectives of these breeds can reach as much as 70 g/day, with birds weighing 2.5 Kg in 35 days [21], average slaughter weight of broilers in the EU and the US is, respectively, 2.5 Kg and 2.9 Kg, reached at 42 and 47 days. This corresponds to an average daily gain (ADG) of 60 and 62 g/day, respectively. We used these values as a benchmark for typical growth rates in fast-growing strains, assuming an ADG of 61 g/day, corresponding to a slaughter weight of 2.5 Kg at 42 days. For the reformed scenario, represented by the use of a slower-growing strain, we assumed an average ADG of 45-46 g/day, hence that the same slaughter weight would be reached in approximately 56 days. This is a growth rate consistent with typical figures achieved by various of the breeds approved under the BCC (e.g. JA 987, 787, Ranger Gold), also referred to as medium- or intermediate-growing broilers. This growth rate also falls within the acceptability of other welfare certification schemes, such as the 'Beter Leven' program in the Netherlands and the 'Deutsche Tierschutzbund' in Germany, which determine a growth rate of up to 45 g/day, and a minimum lifespan of 56 days. Importantly, data on the incidence of important welfare challenges in slower-growing breeds (e.g. proportion of birds with different degrees of lameness) is available predominantly for birds growing at approximately this rate (e.g., [22,23]). The impact of adopting slower-growing breeds with growth rates different than those assumed is, however, discussed in light of the evidence gathered throughout the book.

Key welfare challenges affecting broiler birds are investigated: (1) lameness-causing conditions, which currently affect most fast-growing birds raised commercially, (2) cardiorespiratory insufficiency, particularly pulmonary hypertension leading to ascites and sudden death syndrome, two conditions commonly present in commercial flocks resulting from intense selection for rapid growth; (3) heat stress, regarded as a major challenge among poultry producers, also leading to poor welfare as it triggers physiological and metabolic processes that negatively affect the birds, and (6) the deprivation of potentially motivated behaviors, including foraging, exploration, perching and dustbathing. We also investigate key welfare challenges experienced by broiler breeders, a population that amounts to over 500 million birds (almost twice the total number of cows slaughtered every year). The analysis is focused on the welfare effects of chronic hunger in female breeders (they outnumber males at a ratio of approximately 100 hens to 8 roosters) as a result of severe feed restriction, a pervasive practice in the industry, but also consider the pain associated with cases of egg peritonitis in these hens, a frequent and painful condition that is frequently overlooked. In all scenarios and for all challenges, we conservatively assume that no negative affective state as a result of these conditions is experienced when individuals are sleeping. Box 3 addresses the estimated time birds are awake per day.

It is important to note that, because the main goal of the analysis is to determine the impact of the BCC intervention and genetics on welfare (the absolute difference in the time in pain endured by the average broiler with these reforms), the analysis does not include an exhaustive list of all challenges experienced by broilers, or their breeders, over their lives. For example, we do not analyze the pain associated with many infectious diseases that can affect these birds. Therefore, estimates of cumulative time in pain endured by fast- and slower-growing broilers are underestimated relative to the expected time in pain endured under commercial settings.

### Box 3. Estimated Time Awake per Day

To our knowledge, information on the extent to which birds experience any negative affective states while sleeping is lacking. To err on the side of caution and avoid inflating the estimated times in pain, we conservatively assume that pain is not felt when individuals are sleeping. Overall, sleep in birds is comparable to that in mammals, with both high-amplitude slow-wave and low-voltage fast-wave EEG sleep patterns present in chickens [24]. Chickens have, however, an additional phase of sleep that mammals lack: unihemispheric slow-wave sleep, during which one half of the brain rests while the other half is awake; this is why chickens can sleep with one eye open and the other closed, an adaptation that allows them to be concurrently alert for predators and asleep [25]. Because during this phase fully developed large amplitude slow waves characteristic of electrophysiological sleep may be present [26], we conservatively assume that no awareness of pain is possible even at this time.

Knowledge on the time broilers spend sleeping (as opposed to simply resting) is also scarce. In one study with laying hens, the authors found that the percentage of sleep in a dark period of 10 hours was 83.8%, or slightly over 8 hours [26], as measured by electrophysiological recordings from electrodes implanted in the brain of the chickens. It is thus likely that hens are not continuously asleep during the entire period of darkness. Indeed, active behaviors such as preening during the night have been shown to be common, indicating that when the dark cycle is long birds may spend part of this time awake. Similarly, it is also unlikely that birds will be fully awake for all the time when lights are on. For example, broilers exposed to long cycles of light (20-23 hours per day) spend most of their time lying, and have been shown to doze or sleep for at least a quarter of this time [27]. The duration ranges assumed in the Pain-Tracks presented in this book therefore consider a wide range of hypotheses for the time broiler chickens spend fully awake per day, with minimum and maximum sleeping times of 4 and 10 hours, respectively. Since young chicks may sleep for longer periods, we assume minimum and maximum sleeping times of 8 and 14 hours, respectively, in the first two weeks of life. Given the uncertainty regarding these values, we assume a uniform distribution of probabilities for this range of duration (i.e., all hypotheses are assumed to be equally likely).

## The Next Chapters

This book is laid out as follows. Chapters 2 -5 each investigate the physical and psychological pain experienced by broiler chickens in the conventional and reformed scenarios as a result of the main welfare challenges affecting these birds: lameness-causing conditions, cardiorespiratory insufficiency and the disorders it causes, heat stress and the deprivation of potentially motivated behaviors (foraging/exploration, perching/roosting, dustbathing). Chapter 6 looks at the welfare challenges experienced by broiler breeders. Each chapter starts by presenting a background of the welfare challenge of interest, a review of evidence (empirical data on behavior, preference tests, neurophysiological indicators, homologies with other species and evolutionary reasoning) to infer the duration and intensity of the experience of pain emerging from the challenge, and estimates of the time in pain individuals affected by the harm endure. The prevalence of these welfare offenses in commercial flocks under each scenario is also investigated, as it is necessary to infer their corresponding impacts at the population level.

In Chapter 7, estimates of the impact of the Better Chicken Commitment and use of slower-growing strains on the welfare of broilers at the farm level as a result of the previous ailments are presented and discussed. They are based on the quantification of the total time in pain of different intensities experienced in each scenario, when the cumulative effect of all welfare offenses is considered together. Estimates are accompanied by a discussion of the trade-off between time of life and quality of life in broilers, the potential for welfare improvement solely through management (stocking density, enrichment), as well as the adoption of slower-growing breeds with growth rates different than those assumed. To help inform decisions on research projects and funding, a list of the main research gaps is also provided at the end of each chapter. Estimates are accompanied by a web application (https://pain-track.org/broilers), where it is possible to test the extent to which the estimates are sensitive to the choices and assumptions on the duration, intensity, number of occurrences and prevalence of the welfare challenges under each scenario. All parameters (probabilities, durations, occurrences, prevalence) can be changed, with the immediate update of the results showing the time in each level of pain due to one occurrence of the challenge, the time in pain expected for the average member of the population as a consequence of the challenge (i.e. when its prevalence is considered), and the time in pain expected for the average member of the population as a consequence of all challenges combined.

Finally, in Chapter 8, we quantify the welfare impacts of existing and potential policies on the stunning of poultry prior to slaughter. We use the Cumulative Pain Framework [17] to estimate the loss of welfare endured by birds exposed to electrical waterbath stunning, the most commonly used method for poultry stunning globally [16,17], and multi-stage controlled atmosphere stunning with $CO_2$ the most commonly applied stunning gas under commercial conditions. Here, too, if the reader wants to change any of the assumptions regarding the duration and intensity of the pain depicted in any of the Pain-Tracks presented in Chapter 8, an interactive web application can be used for this purpose, available at https://pain-track.org/broilers/stunning.

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